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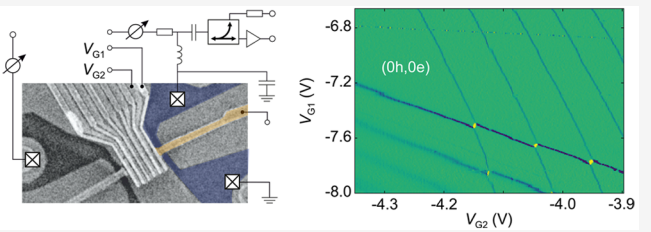
Radio-Frequency Charge Detection on Graphene Electron–Hole Double Quantum Dots

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ABSTRACT: High-fidelity detection of charge transitions in quantum dots (QDs) is a key ingredient in solid-state quantum computation. We demonstrate high-bandwidth radio-frequency charge detection in bilayer graphene quantum dots (QDs) using a capacitively coupled quantum point contact (QPC). The device design suppresses screening effects and enables a sensitive QPC-based charge readout. The QPC is arranged to maximize the readout contrast between two neighboring coupled electron and hole QDs. We apply the readout scheme to a single-particle electron–hole double QD and demonstrate time-resolved detection of charge states as well as magnetic field dependent tunneling rates. This promises a high-fidelity readout scheme for individual spin and valley states, which is important for the operation of spin, valley, or spin-valley qubits in bilayer graphene.



KEYWORDS: *charge detection, bilayer graphene, quantum dot, quantum point contact*

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A fast and state-selective readout method is a prerequisite for operating semiconductor quantum dots (QDs) as hosts for qubits.[1] It is a well-established technique to use capacitively coupled quantum point contacts (QPCs) or single-electron transistors to read out charge transitions in semiconductor QDs.[2,3] Typically, the conductance of the charge detector is measured either directly by transport[4,5] or indirectly via the reflectance of an impedance matched LC resonant circuit connected to the charge detector[6,7] to reach a higher bandwidth.[8] This technique enables time-resolved single-shot detection of charge and spin states in current semiconductor devices.[9−16] Graphene and bilayer graphene (BLG), with their naturally low hyperfine interaction and low spin–orbit coupling, offer a promising new platform for spin- and valley-based qubits.[17−20] BLG is of particular interest thanks to its gate voltage tunable band gap[21] that allows the formation of QPCs and QDs.[22−25] So far, the energy spectra of single-[26,27] and two-particle BLG QD states[28] have been investigated and single-particle spin and valley relaxation times have been studied by standard DC transport techniques.[29,30] Blockade mechanisms of spin and valley states have been demonstrated in BLG double QDs (DQDs)[31−33] as well as a particle-hole symmetry protected spin-valley blockade.[34] In addition, charge sensing has been successfully implemented in BLG QD devices, both by transport measurements of a capacitively coupled QD[35] and by dispersive readout techniques.[36,37] This allowed for time-resolved detection of charge carriers in QDs[38,39] and DQDs[40] in the Hz to low kHz regime. Nevertheless, to study the real-time dynamics of QDs, charge detection with a high single-shot bandwidth is crucial. Here, we present radio-frequency (RF) reflectometry detection of charge states in an electron–hole DQD, via both time-averaged and time-resolved measurements. The RF reflectometry technique allows a bandwidth of up to several MHz. The device has been tailored toward maximum readout contrast between the electron and hole QDs by an in-line arrangement of the QPC with the QDs. We find that the in-plane capacitive coupling can be improved by reducing the charge carrier density between the QDs and the QPC charge detector. We identify the effects of screening and the lateral distance between the QDs and the detector on its sensitivity by comparison with a self-consistent Schrödinger–Poisson solver. Finally, we perform charge detection in the time domain at the interdot transition of a weakly coupled single-particle electron–hole DQD. Thereby, we study the magnetic field dependence of single electron–hole tunneling rates. Time-resolved charge detection of single-particle electron–hole DQDs marks an important step toward the readout of spin and valley qubits in BLG[20] or wide band terahertz photon detection.[41]

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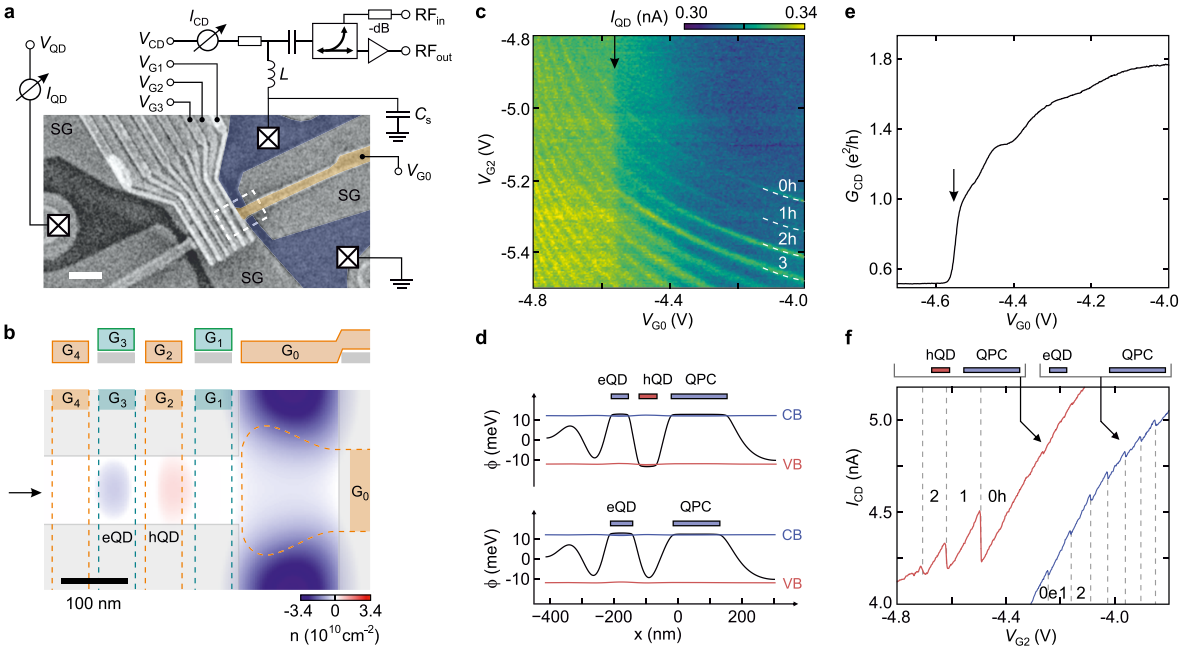


Figure 1. (a) False-colored scanning electron microscopy image of the BLG QD device. Scale bar is 500 nm. Split gates (SGs) define conducting channels; finger gates G1−G3 control the potential along the narrow channel, defining and controlling QDs, and gate G0 (orange) defines a QPC in the wide channel (blue) acting as charge detector. Via three ohmic contacts (⋈), bias voltages can be applied, and current through the device can be measured. One of the contacts is additionally connected to an inductor, L = 3.3 μH, forming an LC resonant circuit together with the stray capacitance of the bond wires, $C_s \approx 0.6$ pF, matching the impedance of the QPC to the 50Ω RF line. For reflectometry measurements, an RF signal (RF_{in}) is applied to the resonant circuit via cryogenic attenuators (−36 dB) and a directional coupler (−20 dB). The reflected signal is amplified (35 dB, Cosmic Microwave CITLF3) at 4 K, followed by additional 28 dB at room temperature. A lock-in amplifier (Zurich Instruments UHFLI 600 MHz) is used for signal generation and homodyne detection. (b) Schematic of the gate structure together with the calculated charge carrier density n induced in the BLG (scale bar is 100 nm). For the calculation the gate voltages were chosen as V_{G1} = −4.44 V, V_{G2} = −4.795 V, V_{G3} = −3.52 V, V_{G4} = −4.54 V, V_{G0} = −4.17 V, V_{SG} = −3.88 V, V_{BG} = 4 V. (c) Current I_{QD} as a function of the gate voltages V_{G0} and V_{G2} with V_{CD} = 250 μV, V_{QD} = 0. Coulomb resonances of a single hole QD formed in the narrow channel close to the T-junction appear. The vertical feature (see arrow) is caused by the formation of a QPC in the wide channel. (d) Potential profiles along the narrow channel (black lines) in between bulk conduction (CB) and valence band (VB) for two different gate voltage configurations defining an electron−hole DQD (upper panel, corresponding to b) and an electron QD (lower panel). (e) Conductance G_{CD} of the charge detector operated in the QPC regime (V_{CD} = 80 μV). Here, the two channels were isolated with G1 and a single hole QD is present in the QD channel (see left schematics in panel (f)). The arrow indicates the operating point set in the following experiments. (f) Jumps in I_{CD} as a function of V_{G2} (see labels and dashed lines) show the formation of single QDs in the few hole (V_{G4} = −4.3 V, red) and few electron (V_{G4} = −5.35 V, blue, offset by −0.5 nA for clarity) regimes (see also corresponding schematics on the top).

The device consists of a heterostructure of BLG encapsulated between two crystals of hexagonal boron nitride (hBN), each ≈50 nm thick, placed on a graphitic back gate (BG). Split gates (SGs, see Figure 1a) are used to form conducting channels. In the wide channel (highlighted in blue in Figure 1a), a QPC beneath the finger gate G0 (orange) acts as a capacitively coupled charge detector. This channel forms a T-junction with a narrow channel beneath the other finger gates (FGs), where the QDs will be localized. This gate architecture is tailored to enhance the capacitive coupling between the detector and the QDs, in particular to optimize the sensitivity to interdot charge transitions. The relative distance of the detector to adjacent QDs is maximized, following designs of other semiconductor QD devices.[13] For details on device design and fabrication, refer to the Supporting Information and to ref 25.

Figure 1b shows a schematic of the top gates at the T-junction (dashed box in Figure 1a) together with the simulated charge carrier density in the BLG for an exemplary set of gate voltages to illustrate how the device is operated. Here, an electron and a hole QD are formed below the FGs, while a QPC is formed below gate G0 at the T-junction. The simulation of a corresponding band edge profile along the center of the narrow channel is shown in the upper panel of Figure 1d. Here, the black curve is the calculated electrostatic potential ϕ varying between the bulk conduction and valence band edges. Importantly, the charge carrier occupations of the individual QDs can be changed by varying the gate voltages, also allowing to form, e.g., just a single electron QD (lower panel in Figure 1d) or a single hole QD (see left schematics in Figure 1f). The simulation is based on a self-consistent Schrödinger−Poisson solver, where a full 3D model of our device was implemented by an adaptive finite element grid with the NGSolve software package[42] (see Supporting Information). The simulations show also that the thickness of the hBN crystals strongly affects the electrostatics of the device. In the case of thin hBN, screening by the close metal gates and the graphite back gate strongly limits the sensitivity of the charge detector (see Supporting Information, Section C).

In the following, the channels between the split gates are set to an n-type polarity by applying a positive voltage to the BG. First, we apply negative voltages V_{G0} and V_{G2} to form an n-type QPC in the wide channel and a hole QD in the narrow channel (see the QPC and hQD in Figure 1b). In Figure 1c, we investigate the configuration of the QPC and just a single

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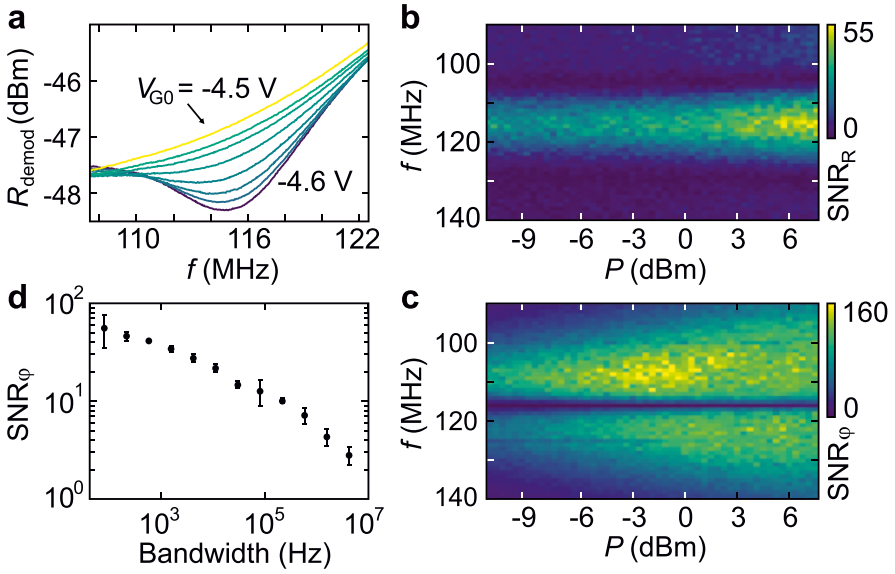


Figure 2. (a) Magnitude of the RF signal as a function of the RF carrier frequency f for different voltages V_{G0} applied to the detector gate (excitation power $P = -10$ dBm). At the resonance frequency f_{res} of the tank circuit, the reflected signal is minimized. (b) Signal-to-noise ratio (SNR) of the magnitude R_{demod} of the RF signal reflected off the charge detector, measured as a function of f and P. The SNR is determined as the ratio of the step height caused by the first charge carrier in the hole QD (0h−1h) and the RMS of the noise. (c) SNR of the phase φ of the RF signal. (d) SNR of the phase response as a function of the measurement bandwidth set by the digital first-order low pass filter of the lock-in amplifier ($P = -11$ dBm).

nearby hole QD (see the left schematic in Figure 1f) by measuring the current through the narrow channel, I_{QD}, as a function of V_{G0} and V_{G2}. Coulomb resonances indicate the formation of a hole QD, the occupation (0h, 1h, ...) of which can be controlled by V_{G2}. With decreasing voltage V_{G0}, the hole QD is dragged toward the charge detector, as shown by the curvature of the Coulomb peaks. To reduce screening of the capacitive coupling between the charge detector and the QDs, G1 is used to deplete the charge carrier density between the QPC and the hole QD (compare with Figure 1b). This results in a low tunnel coupling, where transport through the QD can no longer be measured directly. We then measure the conductance $G_{CD} = I_{CD}/V_{CD}$ of the wide channel as a function of V_{G0}, which is depicted in Figure 1e, where quantized steps indicate the formation of a QPC. The pinch off of the QPC conductance leads to the change in background current in Figure 1c (see arrow). The residual conductance below the pinch off originates from a parallel conductive channel, which is not controlled by the gate voltage V_{G0}, limiting the maximum resistance to 50 kΩ. To optimize the sensitivity of the charge detector, the QPC is tuned to its steepest flank, highlighted by the arrow in Figure 1e, where slight changes in the capacitive environment of the QPC translate into strong changes in its resistance. By measuring the current I_{CD}, the charge occupation of the QDs in the narrower channel can be measured indirectly. Figure 1f shows I_{CD} as a function of V_{G2}. By decreasing V_{G2}, the number of holes occupying the QD under G2 increases, starting from an empty hole QD (red trace in Figure 1f). Each charge transition causes a sharp increase in I_{CD}. When applying a more negative voltage of $V_{G4} = -5.35$ V, a barrier is formed in the channel, which leads to a single electron QD emerging below G3 (see the lower panel in Figure 1d and right schematic in Figure 1f). With decreasing V_{G2}, this electron QD is emptied, as shown by jumps in I_{CD} (blue trace in Figure 1f). Here, the area below G2 acts as the barrier of the electron QD. In this case, the step height of the charge detector response is less pronounced due to the weaker capacitive coupling caused by the larger distance between the electron QD and the QPC charge detector.

To improve the bandwidth and the signal-to-noise ratio (SNR) of the detection signal, we use a radio frequency reflectometry technique. Therefore, we excite the resonance circuit and measure the reflected signal in a homodyne demodulation scheme. Figure 2a shows the magnitude R_{demod} of the demodulated signal as a function of the excitation frequency f for voltages applied to the detector gate between $V_{G0} = -4.6$ V and -4.5 V. The total resistance of the charge detector, including the channel resistance and contact resistances, does not reach the matching condition of $R_{CD} = 100$ kΩ, since it is limited to 50 kΩ (cf. Figure 1e). However, a change of 1.5 dBm can still be observed at the resonance ($f_{res} = 115$ MHz). To quantify the performance of the charge detector using the RF reflectometry technique, we focus on its response to the first hole being loaded to the hole QD (see Figure 1f, transition 0h−1h). The hole QD causes a stronger response in the CD signal than the electron QD because of its smaller lateral distance to the CD. Thus, the SNR provides an upper bound for the CD response to QDs that are farther away. We investigate the SNR, which is defined by $SNR_R = \delta R_{step}/\langle R\rangle$ and $SNR_\varphi = \delta\varphi_{step}/\langle\varphi\rangle$, where δR_{step} and $\delta\varphi_{step}$ are the height of the step induced by the addition of a single charge in the demodulated amplitude (R) or phase (φ). $\langle R\rangle$ and $\langle\varphi\rangle$ are the standard deviations of the signal. Figure 2b shows SNR_R as a function of the excitation frequency f and power P. As expected, SNR_R reaches its maximum at resonance and increases with P. The SNR of the phase, SNR_φ, shows an off-resonance maximum of 160 for P in the range of −3 to 0 dBm (see Figure 2c). This is three times larger than the maximum SNR_R achieved at a larger excitation, $P = 6$ dBm. We attribute the reduced SNR_φ at $P > 0$ dBm to power broadening of the phase response. At high frequencies, SNR_φ scales with the bandwidth as expected in the presence of white noise,

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$SNR_\varphi \propto 1/\sqrt{BW}$, indicating an SNR above unity up to a bandwidth above 10 MHz (see Figure 2d). However, the current setup is limited by the sampling rate of the used RF lock-in amplifier, to 7 MHz.

In addition to the power and frequency of the applied RF carrier, the sensitivity of the charge detector is also influenced by the lateral distance of the QDs and the QPC as well as screening effects due to a finite charge carrier density between them. To investigate these effects, we measure the response of the charge detector on the electron and hole QD charge states using RF reflectometry, while varying the voltage V_{G1} applied to the gate in between the QPC and QDs. The charge stability diagram in Figure 3a shows the derivative of the phase of the reflected signal, $d\varphi/dV_{G2}$, as a function of V_{G1} and V_{G2}. To keep the charge detector at its operating point, we correct for electrostatic cross-talk from gate G1 by varying V_{G0} according to $V_{G0} = -4.737\ V - 0.0193 \cdot V_{G1}$.

The (0h, 0e) region in the upper left of the charge stability diagram (Figure 3a) corresponds to both QDs being empty. Sweeping V_{G1} to more negative values loads holes into the QD between G1 and G2 (Figure 1b), while increasing V_{G2} adds electrons to the neighboring electron QD (see the corresponding labels). The hole QD exhibits a stronger readout contrast than the electron QD, consistent with its closer lateral proximity to the charge detector.

Note that V_{G1} influences the capacitive coupling between the charge detector and the QDs, and hence the detector sensitivity, via two mechanisms: (i) it shifts the lateral position of the QDs, changing their distance to the detector; (ii) it tunes the carrier density in the region between the detector and the QDs, thereby modifying the electrostatic screening. To study this in more detail, we extracted the step height $\delta\varphi_{step}$ as a function of V_{G1} for two distinct charge transitions (white and black dashed lines in Figure 3a). For the (1h, xe) configuration—adding the first hole to the hole QD—we observe that the detector response increases as V_{G1} is made more negative (Figure 3b). In contrast, for the (xh, 2e) configuration—transitioning from one to two electrons in the electron QD—the response increases with more positive V_{G1} (Figure 3c). This charge transition was selected because of its high signal-to-noise ratio. Note that neighboring transitions show a similar qualitative behavior (Supporting Information). To understand these opposing trends, we compare the experimental data with Schrödinger–Poisson simulations of the charge carrier density. The results show that the relative impact of (i) the QPC–QD distance and (ii) screening by intermediate charges depends on the spatial charge distribution. For the (1h, xe) case, the simulations reveal a change only in the amount of charge between the QD and the QPC charge detector (Figure 3d), whereas in the (xh, 2e) case the QPC position also shifts significantly (Figure 3e). Consequently, the simulations reproduce the experimentally observed opposite trends in $\delta\varphi_{step}$ as a function of V_{G1} (Figures 3f,g). This agreement between the experiment and simulation establishes a consistent picture of how V_{G1} controls the detector sensitivity via distance and screening effects.

In the following, we focus on the single-particle electron–hole regime. Figure 4a shows a close-up of the (0h, 0e) ↔ (1h, 1e) charge transition (see Figure 3a), corresponding to the creation and annihilation of electron–hole pairs.[34] Here, the different readout contrast for electrons and holes becomes more apparent. The in-line arrangement of the QDs leads to

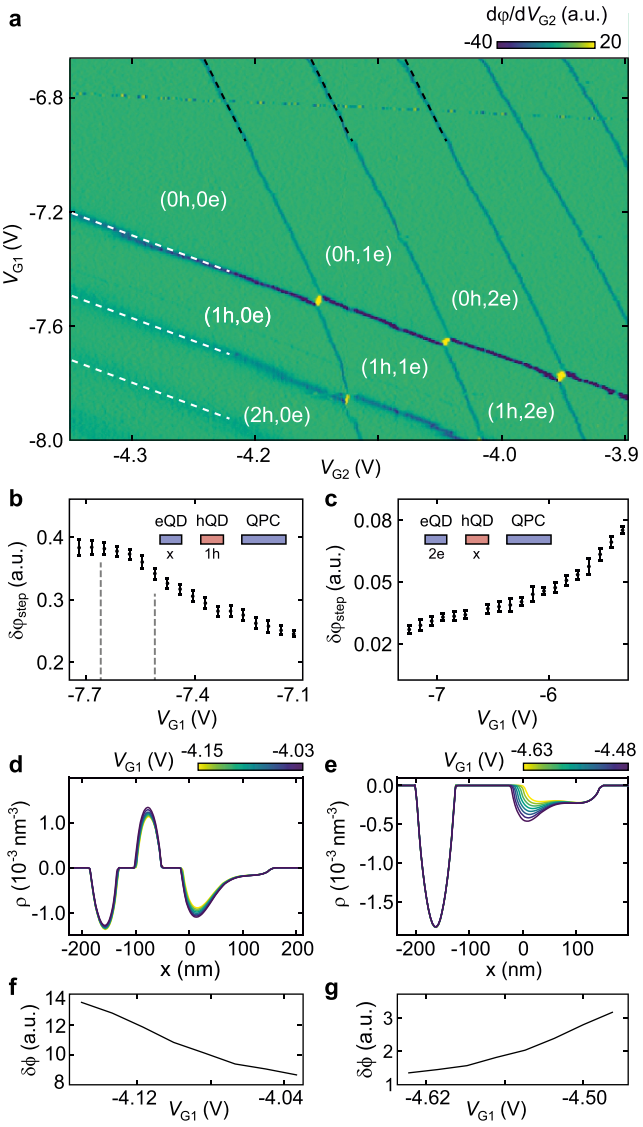


Figure 3. (a) Charge stability diagram of the differential phase $d\varphi/dV_{G2}$ of the reflected RF signal as a function of the gate voltages V_{G1} and V_{G2}. A hole (electron) QD is defined under gate G1 (G2). The charge carrier occupation of an electron (e) and hole (h) QD (see labels) can be tuned down to (0h, 0e). (b, c) Step height $\delta\varphi_{step}$ of the charge detector response at the (b) 0h–1h and (c) 1e–2e charge transition (dashed lines in (a)) as a function of V_{G1}. Charge transitions of the electron QD occur at the gate voltages indicated by dashed lines. The inset depicts the corresponding charge configuration. (d, e) Line cut of the charge carrier density ρ along $y = 0$ in Figure 1b calculated for different V_{G1}. The gate voltages V_{G2}–V_{G4} have been chosen such that (d) the hole QD is occupied by roughly 1 hole and (e) the electron QD contains two electrons. (f, g) Average calculated potential difference $\delta\phi$ at the QPC due to a change in occupation of (f) the hole QD and (g) the electron QD as a function of V_{G1} for the configurations shown in the insets of (b) and (c), respectively.

strong contrast along the interdot transition ($\varepsilon = 0$) and reveals four clearly distinguishable charge states: (0h, 0e), (1h, 0e), (0h, 1e), and (1h, 1e). Figure 4b shows a similar measurement to that in Figure 4a, but with an out-of-plane magnetic field of $B = 1.3$ T applied. The finite magnetic field reduces the interdot tunneling rates, which is reflected in a random telegraph signal (RTS), close to the degeneracy of the (0h, 0e)

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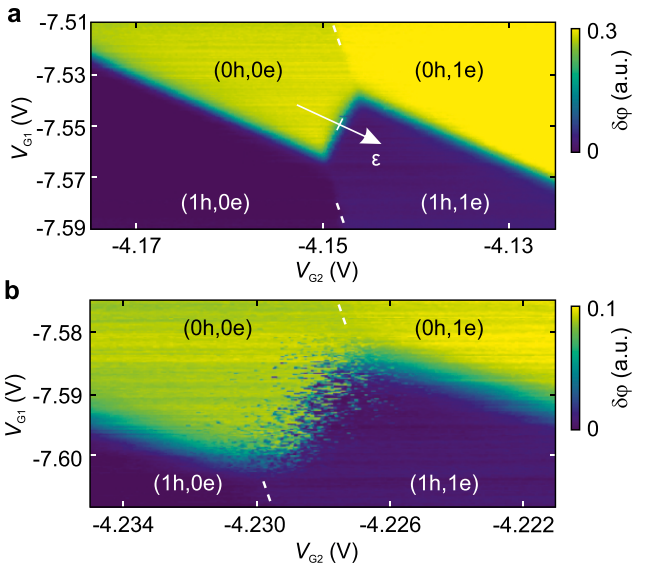


Figure 4. (a) Charge stability diagram showing a close-up of the single-particle electron−hole transition measured in Figure 3a at $B = 0$ T. ε is the detuning energy between the charge states (0h, 0e) and (1h, 1e). A linear background caused by capacitive cross talk between the gate voltages V_{G1} and V_{G2} and the charge detector was subtracted. (b) Same as in (a) but with a perpendicular magnetic field of $B = 1.3$ T applied.

and (1h, 1e) charge states, i.e., at $\varepsilon = 0$, indicating that time-resolved charge detection might be feasible.

To realize time-resolved charge detection, we optimized the charge detector for high-bandwidth operation: electrostatic tuning (see discussion above) combined with RF carrier frequency and amplitude optimization increased the sensitivity and enabled bandwidths in the MHz range—an essential requirement for time-resolved QD state readout in qubit operation.[9,12−14,43,44] We assess the performance of the charge detector for frequencies below the RF carrier frequency of around 115 MHz by measuring the time-dependent detector signal. Therefore, we reduce the tunneling rates by applying an out-of-plane magnetic field of $B = 1.3$ T, similar to the case for Figure 4b. Here, statistical tunnel events can be observed close to the degeneracy of the (0h, 0e) and (1h, 1e) charge states at $\varepsilon = 0$, hinting toward a low interdot tunneling rate. To investigate these processes in detail, we measure the demodulated phase $\delta\varphi$ of the reflected signal as a function of time using a digitizer card, AlazarTech ATS9440. As an example, Figure 5a shows 1 s long traces out of measurements, which took 50 s each. The three traces are measured at different V_{G1}. Individual tunneling events can be observed as an RTS between two levels corresponding to the (0h, 0e) (lower) and the (1h, 1e) (upper) charge configuration. The histograms on the right show the distribution of data points recorded over a time span of 50 s, representing the occupation probability of the two charge configurations. At $\varepsilon = 0$ (red

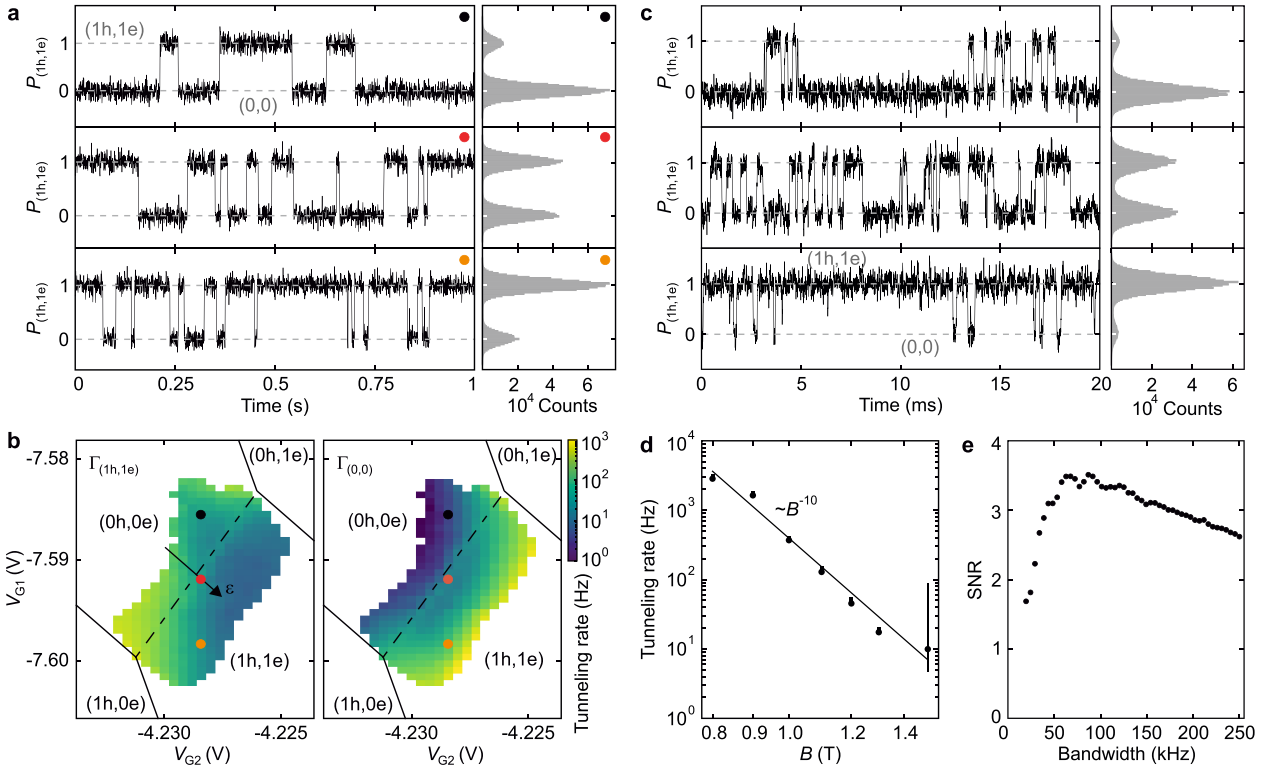


Figure 5. (a) Time-resolved detection of (0h, 0e) ↔ (1h, 1e) charge transitions at $B = 1.3$ T recorded in a time span of 1 s with a sampling rate of 20 kS/s. The data have been low pass filtered at 7.5 kHz. Histograms of the data sets with a trace length of 50 s are shown at the side. (b) Tunneling rates $\Gamma_{(1h,1e)}$ (left) and $\Gamma_{(0,0)}$ (right) for the annihilation and creation of an electron−hole pair, respectively, as a function of V_{G1} and V_{G2} ($B = 1.3$ T). Each data point has been obtained from a single-shot measurement, as in (a). (c) Time-resolved detection in a regime of increased tunnel coupling at $B = 0.9$ T in a time span of 20 ms with a sampling rate of 200 kS/s. The data have been low pass filtered at 77 kHz. Histograms of the data sets with a trace length of 5 s are shown at the side. (d) Combined tunneling rate as a function of the perpendicular magnetic field B, determined at $\varepsilon = 0$. Fitting a power law yields $\Gamma \propto B^{-10.0\pm0.7}$. (e) SNR determined from time-resolved charge detection as a function of the measurement bandwidth ($B = 0.7$ T).

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point in Figure 5a,b), the DQD is found with equal probability in either of the two configurations. For finite detuning ($|\varepsilon| > 0$), the system is predominantly either in the (1h, 1e) configuration (orange point) or in the (0h, 0e) configuration (black point). The corresponding color-coded positions in the charge stability diagram are shown in Figure 5b. We apply a numerical equivalent of a Schmitt trigger[45] thresholding algorithm to detect single tunneling events and determine the time spans $\tau_{(1h,1e)}$ and $\tau_{(0,0)}$ during which the DQD is in the respective state. The rates of the charge transitions (1h, 1e) → (0h, 0e) are $\Gamma_{(1h,1e)} = 1/\langle\tau_{(1h,1e)}\rangle$ and (0h, 0e) → (1h, 1e), $\Gamma_{(0,0)} = 1/\langle\tau_{(0,0)}\rangle$, respectively.[38,46] Figure 5b shows the resulting rates $\Gamma_{(1h,1e)}$ (left) and $\Gamma_{(0,0)}$ (right) as a function of V_{G1} and V_{G2} at $B = 1.3$ T. At $\varepsilon < 0$, $\Gamma_{(1h,1e)}$ is increasing while $\Gamma_{(0,0)}$ is decreasing and vice versa for $\varepsilon > 0$. This result matches the RTS measurements in Figure 5a. At $\varepsilon = 0$, we find equal transition rates of $\Gamma_{(1h,1e)} = \Gamma_{(0,0)} \approx 20$ Hz. Here, resonant tunneling between the electron and hole QD as well as co-tunneling from the reservoirs could contribute to the transition between the charge states.[47] We decrease the magnetic field to $B = 0.9$ T to increase the interdot coupling. Figure 5c shows 20 ms long time traces that are taken from a measurement recorded for 5 s at different positions along the interdot transition (compare with Figure 5a,b). Here, the tunneling events appear on time scales that are around 100 times faster than at $B = 1.3$ T (see Figure 5a). To capture these events, the low pass cutoff frequency and the sampling rate of the Alazar card had to be increased to 77 kHz and 200 kS/s, respectively, leading to a decrease of the SNR. To quantify the effect of magnetic field tunable tunneling rates, which is in qualitative agreement with previous work,[28,33,48,49] we measure the transition rates between the charge states (0h, 0e) and (1h, 1e) at $\varepsilon = 0$ as a function of B. The result is depicted in Figure 5d and shows a strong dependency, following a power law of approximately $\Gamma \propto B^{-(10.0\pm0.7)}$. By fitting a double Gaussian function to the histograms, which are exemplarily shown in Figures 5a and b, the SNR can be extracted from the time-dependent measurements.[50] The corresponding data as a function of detection bandwidth is depicted in Figure 5e (here, exemplary for $B = 0.7$ T). We found that an SNR above 2.5 is maintained for a bandwidth exceeding 250 kHz.

In summary, we demonstrate RF reflectometry charge detection of an electron−hole DQD in BLG using a capacitively coupled QPC. In average signal measurements of the charge detector response, we reach a maximum SNR of 160 and a bandwidth of up to 7 MHz, which exceeds the bandwidth of 10 kHz measured with previous two-channel devices[27] by 3 orders of magnitude. So far, comparable values of a bandwidth of up to 10 MHz were achieved by dispersive sensing using a microwave resonator.[37] Improving the matching condition between the resonant circuit and the QPC can further increase the sensitivity of the charge detector. The sensitivity can be tuned electrostatically by using the finger gate between the detector and QDs. We identify screening effects and the distance of the detector and QDs as the relevant mechanisms limiting the sensitivity by comparison with a model based on a self-consistent Schrödinger−Poisson solver. Finally, we demonstrate the distinction of the charge states of a single-particle electron−hole DQD in the time domain with a measurement bandwidth of up to 250 kHz, exceeding state-of-the-art experiments on electron QDs in BLG, which reach values of 1−10 kHz.[38−40] With this technique, we determine magnetic field dependent single electron−hole tunneling rates. Importantly, the electron−hole DQD in BLG is a particle-hole symmetric system with a robust single-particle spin and valley blockade.[34] This mechanism, together with the presented time domain measurements, may allow for high fidelity spin-to-charge and valley-to-charge conversion, providing a reliable read-out scheme for spin and valley states.[20] This paves the way for measurements of the relaxation and coherence times of spin and valley states, marking a milestone toward the realization of spin, valley, and spin−valley qubits. Beside qubits, further applications of time-resolved readout of electron−hole DQDs in BLG may include wide band single-photon THz detectors,[41,51] electron−hole pair pumps,[52] and Cooper pair splitters.[53]

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are openly available in a Zenodo repository: DOI 10.5281/zenodo.17816134.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04648.

Details on the device design, on the Schrödinger−Poisson model, and on the impact of the insulator thickness (PDF)

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[▽]K.H. and S.M. contributed equally to this work. K.H., S.M., C.V., and C.S. conceived this experiment. K.H., S.M., H.D., S.R., and T.D. fabricated the device. K.H., S.M., and H.D. performed the measurements and analyzed the data with the help of L.V. and L.S. Ş.D. and F.L. performed the electrostatic simulations of the device architecture. K.W. and T.T. synthesized the hBN crystals. C.V. and C.S. supervised the project. K.H., S.M., H.D., C.V., Ş.D., F.L., and C.S. wrote the manuscript with contributions from all authors.

Notes

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